Example 10.11

If 2% of electric bulbs manufactured by a certain company are defective find the probability that in a sample of 200 bulbs (i) less than 2 bulbs are defective (ii) more than 3 bulbs are defective. [e-4 = 0.0183]

Solution:

Let X denote the number of defective bulbs

$X \sim P(\lambda)$

$$\therefore \quad P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} \qquad x = 0, 1, 2, \ldots \infty$$

Given $p = P$ (a defective bulb) $= 2\% = \frac{2}{100} = 0.02$

$$n = 200$$

$$\therefore \quad \lambda = np = 200 \times 0.02 = 4$$

$$\therefore \quad P(X = x) = \frac{e^{-4}4^x}{x!}, \quad x = 0, 1, 2, \ldots \infty$$

(i) P (less than 2 bulbs are defective)

$$= P(X < 2)$$

$$= P(x = 0) + P(x = 1)$$

$$= \frac{e^{-4}.4^0}{0!} + \frac{e^{-4},4^1}{1!}$$

$$= e^{-4}(1 + 4)$$

$$= 0.0183 \text{ x } 5$$

$$= 0.0915$$

(ii) P (more than 3 defectives)

$$= P(X > 3)$$

$$= 1 - P(X \leq 3)$$

$$= 1 - \{P(x = 0) + P(x = 1) + P(x = 2) + P(x = 3)\}$$

$$= 1 - \left\{\frac{e^{-4}.4^0}{0!} + \frac{e^{-4}.4^1}{1!} + \frac{e^{-4}.4^2}{2!} + \frac{e^{-4}.4^3}{3!}\right\}$$

$$= 1 - e^{-4}\{1 + 4 + 8 + 10.667\}$$